02 / ALGEBRA / CLASS_17_VECTORS / EX26

# Engineering and mathematics theory guide

Dot product, angles and the mathematical meaning of mechanical work.

**DOCUMENT 01**

## Learning objectives

- Define and use the dot product through components.
- Derive its relationship to angle and orthogonality.
- Separate algebraic calculations from physical work.

**Prerequisites:** Prerequisites: EX25 completed, NumPy arrays, vector components and Euclidean norm.

Reading order: 01 Theory → 02 Reference → 03 Exercise → 04 Pseudocode → Coding.

## 1. Objective: from algebra to engineering

The dot product is an operation between vectors that returns a **scalar number**. We first study vectors without units, then its geometric meaning, and finally its application to mechanical work.

## 2. Algebraic definition

For two vectors with the same dimension, the dot product multiplies matching components and adds these products. It is neither vector addition nor a componentwise vector-valued product.

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i$$

$$\mathbf{u} = (u_x, u_y), \quad \mathbf{v} = (v_x, v_y)$$

$$\mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y$$

In three dimensions, we simply add one more component:

$$\mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y + u_z v_z$$

> The result is a scalar. Both vectors must have the same number of components. $u_i$ and $v_i$ are scalar components.

## 3. Key properties

The dot product is commutative, distributive and compatible with scalar multiplication:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

$$\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$$

$$(\alpha \mathbf{u}) \cdot \mathbf{v} = \alpha (\mathbf{u} \cdot \mathbf{v})$$

One important property connects the dot product with the 2-norm studied in EX25:

$$\mathbf{u} \cdot \mathbf{u} = \sum_{i=1}^{n} u_i^2 = \|\mathbf{u}\|_2^2$$

$$\|\mathbf{u}\|_2 = \sqrt{\mathbf{u} \cdot \mathbf{u}}$$

Therefore, the dot product of a vector with itself cannot be negative and is zero only for the zero vector.

## 4. Worked mathematical example

Given these vectors, with no units:

$$\mathbf{u} = (2, 1), \qquad \mathbf{v} = (1, 3)$$

Expand the dot product component by component:

$$\mathbf{u} \cdot \mathbf{v} = 2(1) + 1(3)$$

$$\mathbf{u} \cdot \mathbf{v} = 5$$

We also calculate both vector norms:

$$\|\mathbf{u}\|_2 = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\|\mathbf{v}\|_2 = \sqrt{1^2 + 3^2} = \sqrt{10}$$

## 5. Geometric meaning: the angle

There is an equivalent expression for the dot product using the angle θ between two nonzero vectors:

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos\theta$$

We can solve for the angle provided neither norm is zero:

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}$$
$$\theta = \arccos\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}\right)$$

For our previous example:

$$\cos\theta = \frac{5}{\sqrt{5}\sqrt{10}} = \frac{1}{\sqrt{2}}$$
$$\theta = 45^\circ$$

### 5.1 Where does the geometric formula come from?

It follows from the norm of the difference and the law of cosines. Expanding the squared norm:

$$\|\mathbf{u} - \mathbf{v}\|_2^2 = (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v})$$
$$= \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2\mathbf{u} \cdot \mathbf{v}$$

The law of cosines describes that same length as:

$$\|\mathbf{u} - \mathbf{v}\|_2^2 = \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2\|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos\theta$$

Equating both expressions gives the geometric definition of the dot product.

## 6. Sign and orthogonality

A positive dot product corresponds to an acute angle; zero means perpendicular vectors if both are nonzero; a negative dot product corresponds to an obtuse angle. This describes their relative orientation.

$$\mathbf{u} \cdot \mathbf{v} > 0 \Rightarrow \theta < 90^\circ$$
$$\mathbf{u} \cdot \mathbf{v} = 0 \Rightarrow \theta = 90^\circ$$
$$\mathbf{u} \cdot \mathbf{v} < 0 \Rightarrow \theta > 90^\circ$$

For example, (2, 1) · (1, −2) = 2 − 2 = 0. These two nonzero vectors are orthogonal.

> Important: the zero vector has zero dot product with every vector, but its angle is undefined. You cannot compute an angle by dividing by its norm.

## 7. Cauchy–Schwarz inequality

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|_2 \|\mathbf{v}\|_2$$

This inequality guarantees that the quotient used as a cosine lies mathematically between −1 and 1 (for nonzero vectors). Floating-point rounding can slightly exceed this range; we will use np.clip() to guard

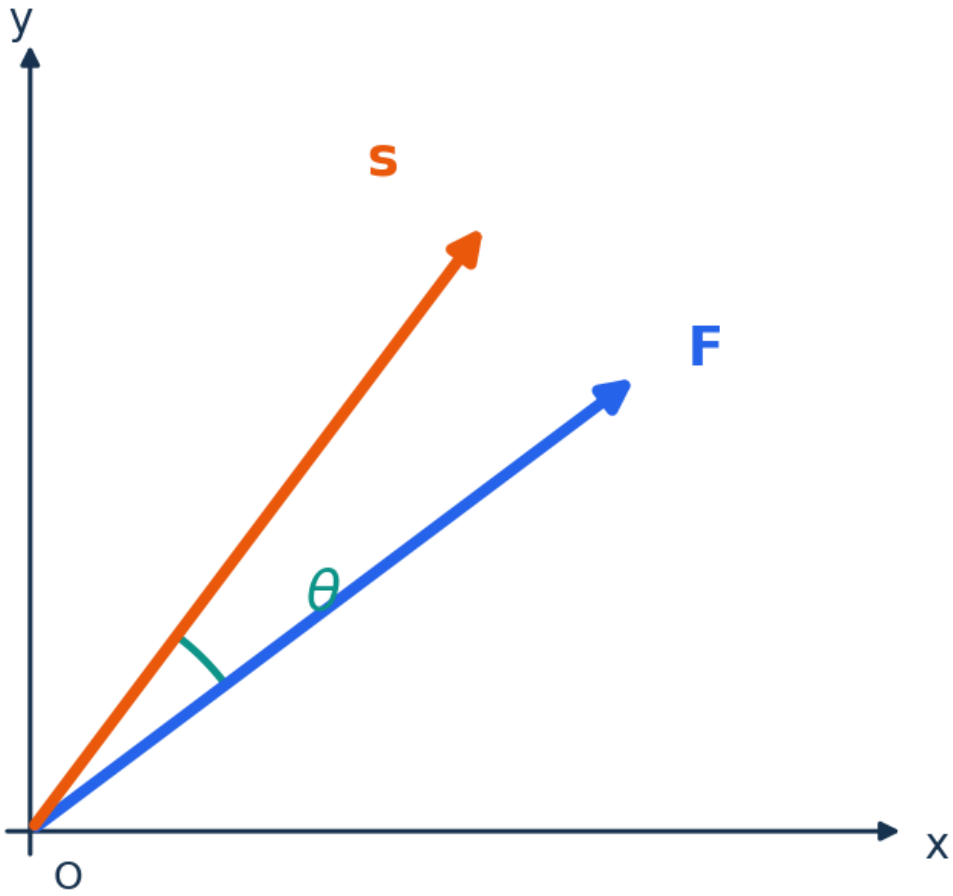
against it.

## 8. Physics application: mechanical work

For a constant force **F** during a displacement **s**, mechanical work is the dot product of those quantities. We are not adding forces: we multiply force by displacement:

$$W = \mathbf{F} \cdot \mathbf{s} = \|\mathbf{F}\|_2 \|\mathbf{s}\|_2 \cos\theta$$

Units are newton times meter: N·m = J (joules). The sign of W indicates whether the force assists, is perpendicular to, or opposes displacement.




Direction sketch: drawn lengths do not compare force (N) with displacement (m).

Engineering example: force and displacement in Cartesian coordinates:

$$\mathbf{F} = (8, 6)\ \text{N}, \quad \mathbf{s} = (3, 4)\ \text{m}$$
$$W = 8(3) + 6(4)$$
$$W = 48\ \text{J}$$

Check the geometric interpretation:

$$\|\mathbf{F}\|_2 = 10\ \text{N}, \quad \|\mathbf{s}\|_2 = 5\ \text{m}$$
$$\cos\theta = \frac{48}{10(5)} = 0.96$$

$$\theta \approx 16.26^\circ$$

A force perpendicular to displacement does zero work; a force opposing displacement does negative work. This interpretation does not depend on the chosen coordinate axes.

## 9. Connection to Python

Once the operation is understood, NumPy represents vectors with np.array(), computes the product with np.dot(), norms with np.linalg.norm(), and angles with np.arccos() and np.degrees(). The Reference Guide describes each function and its inputs.

> Summary: the dot product returns a number; its value depends on both lengths and relative direction. It is used to compare directions, detect orthogonality and compute mechanical work.